YANG Hao

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EDUCATION

Northwestern Polytechnical University (NPU) | Xi'an, China Aug. 2022 - Jul. 2026 (Expected)

- Program: **Carbon Neutrality and Smart Energy Innovation Class (Elite Program, Top 20% high school graduates by province)**, Honors College
- Major: Computer Science and Technology (Bachelor of Engineering)
- Minor: Smart Energy
- GPA: **3.739/4.1**

The Chinese University of Hong Kong, Shenzhen | Shenzhen, China Jul. 2025 - Aug. 2025

- Program: the 2025 SSE, CUHK-Shenzhen Elite Undergraduate Summer Camp

University of Oxford | Oxford, United Kingdom Jul. 2024 - Aug. 2024

- Program: Summer School Program, STEM
- Grade: **A**

University of British Columbia | Kelowna, Canada Jul. 2023 - Aug. 2023

- Program: Summer Program, Engineering
- Grade: **Gold Medal**

INTERNSHIP EXPERIENCE

Text-Guided Remote Sensing Super-Resolution Network with Multi-Stage Tail Refinement
Nov. 2025 - Dec. 2025

Supervised by Prof. Man-On Pun, CUHK(SZ)

- *Outcome*: Wrote a paper named **Text-Guided Remote Sensing Super-Resolution Network with Multi-Stage Tail Refinement**, which is expected to be submitted to IEEE TGRS.
  - ✓ Coupled a tailless multimodal remote sensing super-resolution backbone with a lightweight, multi-stage Tail refinement strategy.
  - ✓ Introduced a residual-pretraining-and-fine-tuning paradigm to train the tail model to enhance the remote sensing super-resolution task.

Text-Conditioned Semantic Efficient Remote Sensing Image Transfer System Jul. 2025 - Aug. 2025

Supervised by Prof. Man-On Pun, CUHK(SZ)

- *Outcome*: Wrote a paper named **Text-RSR: Semantic-Aware Remote Sensing Image Transmission with Text-Guided Reconstruction**, which is expected to be submitted to CVPR 2026.
  - ✓ Proposed a semantic-efficient RS image transfer framework, enabling low-bandwidth and information-preserving transmission via LR–text pairs.
  - ✓ Developed a dual-head network that extracts fine-grained local visual features from images while simultaneously preserving the global textual and semantic representations captured by CLIP.

RESEARCH EXPERIENCE

Research Program on Remote Sensing Change Detection Jul. 2024 - Apr. 2025

Supervised by Prof. Zhiyu Jiang, NPU

- *Outcome*: Published a paper entitled **Multimodal Difference Augmentation Learning for Remote Sensing Change Detection** in IEEE Transactions on Geoscience and Remote Sensing.
  - ✓ Introduced an MdaCD model which achieved state-of-the-art performance with IoU scores of 84.88% on LEVIR-CD and 71.96% on SYSU-CD.
  - ✓ Employed a CLIP-Guided Masking process to conceal regions that are highly unlikely to change and

subsequently generated text prompts.
    - ✓ Developed a Multimodal Fusion and Difference Augmentation process to enhance the discrepancy between visual and textual information, facilitating differential feature learning and improving the accuracy of change detection.

COMPETITIONS

The 2023 Mathematical Contest in Modeling (MCM) Feb. 2023

Competitor, Modeler | ***Awarded: Honorable Mention***

- ➢ Background: Regularity Behind Random Puzzles, three tasks to be addressed.
- ➢ Completed Tasks 2 and 3, including problem interpretation, algorithm design, code implementation, and contest paper writing.
- ➢ Task 2: Employed a Binary Decision Tree model to analyze the features of the given word and predict the distribution of attempts required to guess the word (1 to 6 attempts or failure).
- ➢ Task 3:
    - ✓ Adopted the K-means++ algorithm to classify words into two categories: "simple" and "difficult."
    - ✓ Employed a Binary Logistic Regression model to predict the difficulty of words based on the differences in letter frequency between the two categories.

The 2021 ICPC China Shaanxi Provincial Programming Contest Oct. 2022

Competitor, Modeler | ***Awarded: Bronze Medal***

- ➢ Background: Responsible for Problems F and G and collaborated with team members to address Problem B. The Bronze Medal was awarded to the team for completing three Problems (B, G, J).
- ➢ Problem F: attempted to optimize a brute-force enumeration algorithm but failed to resolve the Problem within the competition time.
- ➢ Problem G: utilized mathematical modeling to calculate the maximum water storage using the arithmetic series sum formula, successfully solving the Problem.
- ➢ Problem B: participated in the process by transforming the complex constraints into a mathematical model and using a Binary Indexed Tree (BIT) for point updates and range queries.

PUBLICATION

- ➢ **Hao Yang,** Zhiyu Jiang, Dandan Ma, Qi Wang, "Multimodal Difference Augmentation Learning for Remote Sensing Change Detection", *IEEE Transactions on Geoscience and Remote Sensing*. DOI: 10.1109/TGRS.2025.3630710.

ADDITIONAL INFORMATION

- ➢ Award and Scholarship:
    - ✓ The 2023 Xiaomi Excellence Award
    - ✓ The 2023 First Class Scholarship (University-level)
- ➢ **Programming: C++, Python, MATLAB, LaTeX, Markdown**
- ➢ Qualification: Certified Software Professional (CSP), scored 345/500 (Top 2% among all participants)
- ➢ Language Proficiency:
    - ✓ English: Fluent (IELTS 7.5, GRE Verbal 150/Quant 167)
    - ✓ Mandarin: Native
    - ✓ Cantonese: Fluent